

[I225] Statistical Signal Processing(E) Office Hour 3

1. Consider a discrete-time LTI system used to create a simple echo effect. The system has the following impulse response

$$h[n] = \delta[n] + 0.4\delta[n - 2]$$

The input signal is:

$$x[n] = \{1, 2, 0, -1\}$$

where the first element corresponds to $n = 0$.

- (a) Explain what the impulse response means physically.
- (b) Calculate the output signal $y[n] = x[n] * h[n]$.

Answer:

(a)

The impulse response contains two parts.

The term $\delta[n]$ means that the original signal passes through the system without delay.

The term $0.4 \delta[n - 2]$ means that an attenuated copy of the signal appears after 2 samples with gain 0.4.

(b)

Since the system is LTI, the output is given by convolution:

$$y[n] = x[n] * h[n] = x[n] + 0.4 x[n - 2]$$

Using $x[n] = 0$ outside the given range:

$$y[0] = x[0] + 0.4x[-2] = 1$$

$$y[1] = x[1] + 0.4x[-1] = 2$$

$$y[2] = x[2] + 0.4x[0] = 0.4$$

$$y[3] = x[3] + 0.4x[1] = -1 + 0.8 = -0.2$$

$$y[4] = 0 + 0.4x[2] = 0$$

$$y[5] = 0 + 0.4x[3] = -0.4$$

So the final output is:

$$y[n] = \{1, 2, 0.4, -0.2, 0, -0.4\}$$

2. Consider a continuous-time linear time-invariant system with impulse response:

$$h(t) = e^{-2t}U(t)$$

The input random process is:

$$X(t) = f(t) + v(t), \quad f(t) = e^{-t}U(t)$$

where $v(t)$ is a zero-mean random process, so $E\{v(t)\} = 0$.

(a) Find the mean of the input process $\eta_X(t)$.

(b) Find the mean of the output process $\eta_Y(t)$.

Answer:

(a)

The mean of the input process is:

$$\eta_X(t) = E\{X(t)\} = E\{f(t) + v(t)\} = f(t) + E\{v(t)\}$$

$$\eta_X(t) = e^{-t}U(t)$$

(b)

For a linear system, the expectation and the linear operation can be interchanged:

$$E\{L[X(t)]\} = L\{E\{X(t)\}\}$$

Therefore, the output mean is the response of the system to the input mean:

$$\eta_Y(t) = \eta_X(t) * h(t)$$

For $t \geq 0$:

$$\eta_Y(t) = \int_0^t e^{-(t-\alpha)} e^{-2\alpha} d\alpha$$

$$\eta_Y(t) = e^{-t} \int_0^t e^{-\alpha} d\alpha$$

$$\eta_Y(t) = e^{-t}(1 - e^{-t})$$

So the final answer is:

$$\eta_Y(t) = e^{-t}(1 - e^{-t})U(t)$$

3. Consider a zero-mean white noise process $W(t)$ injected into an LTI system at $t = 0$. The input autocorrelation is:

$$R_{WW}(\tau) = q\delta(\tau)$$

The impulse response of the system is:

$$h(t) = e^{-ct}U(t), \quad c > 0$$

- (a) Find the output power $E\{Y^2(t)\}$.
- (b) Find the steady-state output power as $t \rightarrow \infty$.

Answer:

(a)

Since the white noise is injected at $t = 0$, the effective input power starts from zero and then passes through the system. For white noise input with autocorrelation $q\delta(\tau)$, the output power is obtained from the squared impulse response:

$$E\{Y^2(t)\} = q \int_0^t h^2(\alpha) d\alpha$$

Substituting $h(t) = e^{-ct}U(t)$:

$$E\{Y^2(t)\} = q \int_0^t e^{-2c\alpha} d\alpha$$

$$E\{Y^2(t)\} = q \left[\frac{1 - e^{-2ct}}{2c} \right]$$

So the output power is:

$$E\{Y^2(t)\} = \frac{q}{2c} (1 - e^{-2ct})$$

(b)

When $t \rightarrow \infty$, $e^{-2ct} \rightarrow 0$. Therefore, the steady-state output power is:

$$\lim_{t \rightarrow \infty} E\{Y^2(t)\} = \frac{q}{2c}$$

4. Suppose $X(t)$ is a wide-sense stationary random process with autocorrelation function:

$$R_{XX}(\tau) = \sigma^2 e^{-a|\tau|}, \quad a > 0$$

- (a) Find the power spectral density $S_{XX}(\omega)$.
- (b) Verify the signal power from the autocorrelation function.

Answer:

According to the Wiener-Khinchin theorem, the power spectral density is the Fourier transform of the autocorrelation function:

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-j\omega\tau} d\tau$$

Substituting $R_{XX}(\tau) = \sigma^2 e^{-a|\tau|}$:

$$S_{XX}(\omega) = \sigma^2 \int_{-\infty}^{\infty} e^{-a|\tau|} e^{-j\omega\tau} d\tau$$

Using the standard Fourier transform pair:

$$e^{-a|\tau|} \Leftrightarrow \frac{2a}{a^2 + \omega^2}$$

Therefore:

$$S_{XX}(\omega) = \frac{2a\sigma^2}{a^2 + \omega^2}$$

The signal power is obtained from $R_{XX}(0)$:

$$P = E\{X^2(t)\} = R_{XX}(0) = \sigma^2$$

This is also consistent with the PSD integral:

$$P = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega = \sigma^2$$

5. Consider a zero-mean white noise process $X(t)$ with power spectral density:

$$S_{XX}(\omega) = \frac{N_0}{2}$$

The process passes through an ideal low-pass LTI system with frequency response:

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & |\omega| > \omega_c. \end{cases}$$

(a) Find the output power spectral density $S_{YY}(\omega)$.

(b) Find the total output power.

Answer:

(a)

For a wide-sense stationary input passing through an LTI system, the output power spectral density is:

$$S_{YY}(\omega) = |H(j\omega)|^2 S_{XX}(\omega)$$

Since the input is white noise, $S_{XX}(\omega) = N_0/2$. The ideal low-pass filter only keeps frequency components within $|\omega| \leq \omega_c$.

$$S_{YY}(\omega) = \begin{cases} \frac{N_0}{2}, & |\omega| \leq \omega_c, \\ 0, & |\omega| > \omega_c. \end{cases}$$

(b)

The total output power is the integral of the output PSD:

$$P_Y = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega) d\omega$$

$$P_Y = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{N_0}{2} d\omega$$

$$P_Y = \frac{1}{2\pi} \left(\frac{N_0}{2} \right) (2\omega_c)$$

So the final answer is:

$$P_Y = \frac{N_0 \omega_c}{2\pi}$$

This means that the low-pass system converts ideal white noise into band-limited noise.